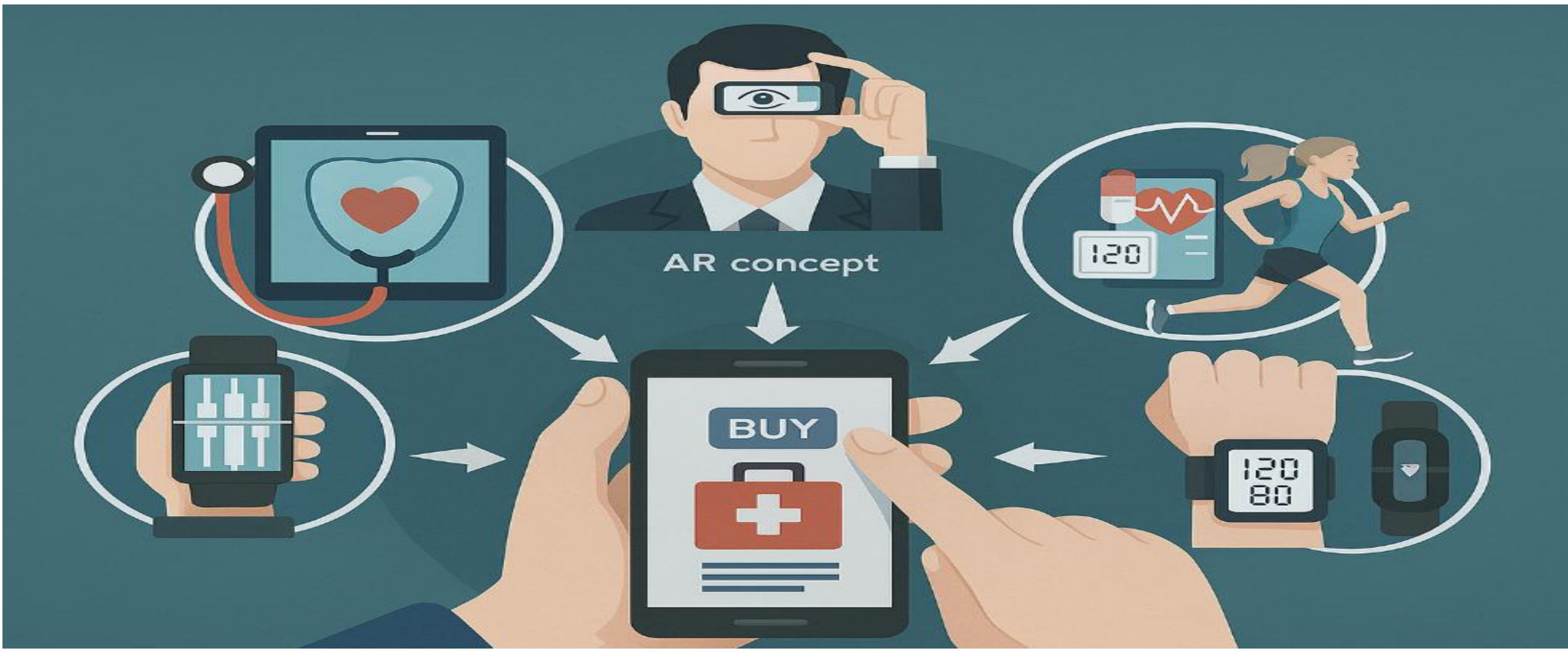
Executive Summary

This business report presents a comprehensive requirements analysis and system modelling solution for Horizon Health's proposed mobile application, prepared by team of business analysts for Dr. Olivia Carter (Owner) and Mr. James Wilson (IT Director). The app is designed to support Horizon Health's franchise model by streamlining lead management, service coordination, and invoicing through a secure, AI ready mobile platform.

The report employs structured modelling methods including UML Use Case, Sequence, and Class Diagrams, along with a BPMN process model, to map out key workflows from lead assignment to service delivery and invoicing. Franchisees are the primary users, supported by administrative roles for training, compliance, and operational control.

A notable innovation is the integration of Augmented Reality (AR) with generative AI to enhance patient care. Features such as Scan Medication via AR and View AR Exercise Guide empower clients with real time, interactive guidance for medication adherence and rehabilitation.

Findings confirm the feasibility of a modular, AI enhanced system that can automatically allocate clients to qualified franchisees and generate invoices upon client request, reducing dependency on Horizon Health's central team. The solution offers scalable efficiency, improved care coordination, and supports Horizon Health's digital transformation and growth objectives.



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1. Section 1

1.1 Use Case Diagram

Table 1 below outlines the high-level requirements for each key actor involved in the Horizon Health mobile app, capturing the essential capabilities needed to support their respective roles in the system.

Table 1: High Level User Requirements

Actor	High-Level Requirement
Franchisee	Must be able to register, verify identity, manage profile, and convert leads to clients before initiating service or invoicing.
Client	Must be able to receive service and digital invoices, and view AR medication and exercises.
Call Centre Administrator	Must be able to assign, cancel, or transfer leads based on franchisee requests or capacity.
System	Must support lead conversion, service scheduling, invoice generation, and reporting capabilities.
External Accounting System	Must integrate with the app for automated income/expense reconciliation and audit readiness.
Operations Manager	Must restrict franchisees to only receive leads for services they are registered to provide.
Marketing Manager	Must be able to generate performance and revenue reports based on service and invoice data.
Training Department	Must be able to update training content to support franchisee queries and onboarding.
Accounts Department	Must generate and issue monthly invoices to franchisees based on brand usage and lead allocation.
AI Analytics Engine	Must analyse franchisee performance metrics to identify operational improvements and trends.

Table 2 captures the outcomes of the requirements elicitation process conducted for Horizon Health's proposed mobile application. It outlines critical insights into the system's purpose, scope, functional expectations, constraints, and stakeholder considerations. These insights form the foundation for system modelling and guide the development of user centred functionalities, ensuring alignment with operational goals, regulatory needs, and franchisee workflows.

Table 2: Requirements Elicitation

Sections	Questions to investigate / Horizon Health Answers
Purpose of the system	To enhance in-home care service delivery and operational efficiency through a mobile platform for patients and franchisees.
Scope of the system	Franchisees and patients will use the system daily. Expected growth includes franchise expansion across regions. It affects service coordination, invoicing, lead management, and training. Interfaces with external accounting and potential AI tools.
Management objectives	Improve visibility of franchisee operations, enable service tracking, support franchise business model, generate financial and operational insights.
Functions	Enable lead management, registration, client conversion, service updates, invoicing, territory preference settings, report generation, training resource access, and accounting integration.
Constraints	Must meet healthcare data privacy, security (possibly HIPAA or local equivalents), deadlines for app launch, integration constraints with external systems, and ATO compliance. Real-time updates required.
Additional user resource requirements	Franchisees may require digital onboarding support; no additional physical resources explicitly mentioned but customer support might scale.
Assumptions	Franchisees must register with Horizon email and verify ID. AI and external accounting integration are assumed to be backend-supported.
Open issues	Clarify scope of AI usage and accounting integration, and rules around territory preference enforcement.
Participation list	Project team, business analysts, franchisee reps, training team, marketing, operations, and accounting stakeholders.

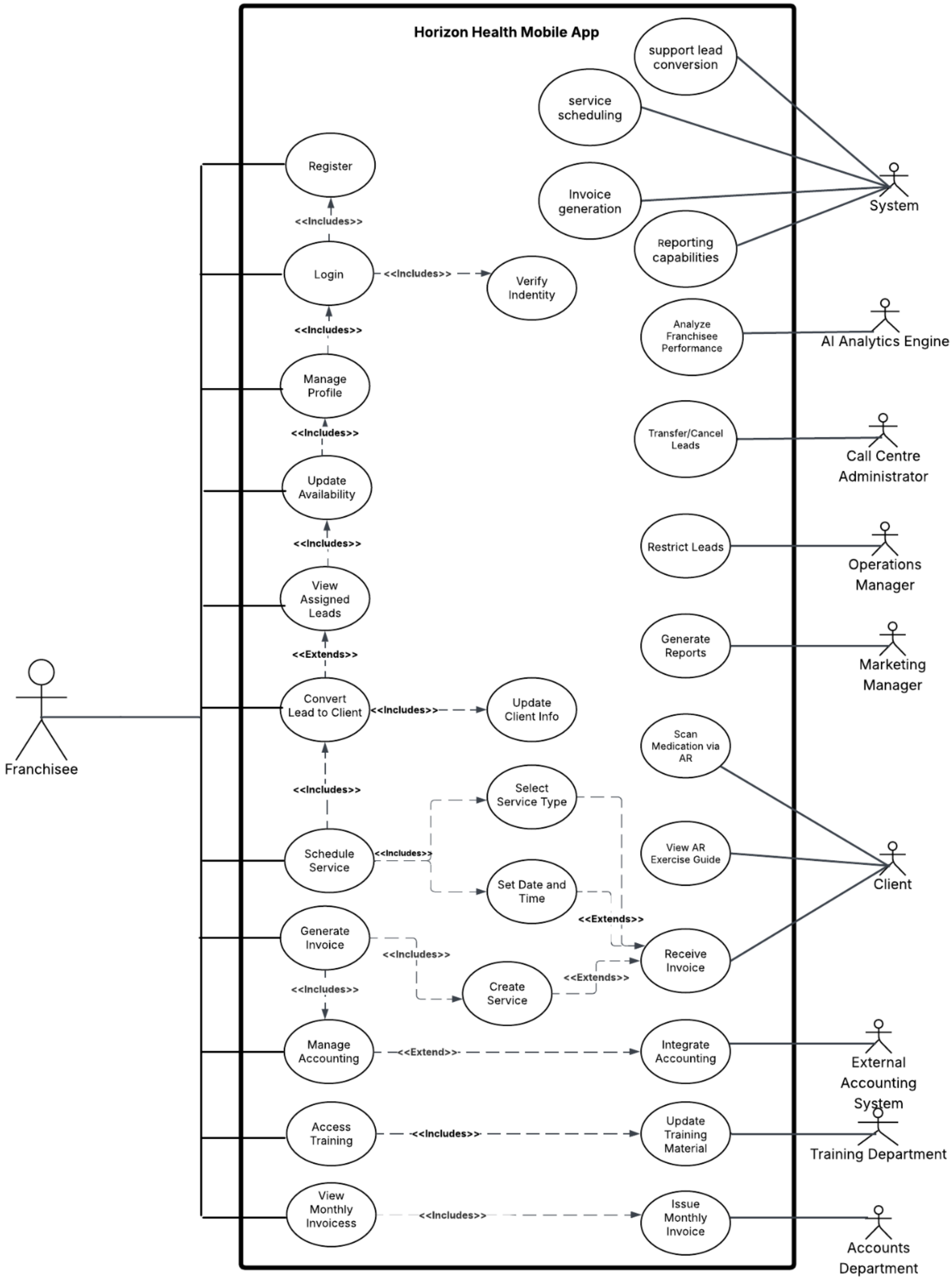


Figure 1: Use Case Diagram

Actors interacts with the system based on their role specific needs, and their actions are reflected through distinct use cases enclosed within the system boundary of the mobile app (Al-Fuqaha et al., 2015).

Primary Actor: Franchisee

- Performs most operational activities in the system.
- Initiates interaction through the Login process, which includes Register.
- The login flow incorporates a mandatory Verify Identity step to ensure secure authentication.
- After verification, the Franchisee can access the Manage Profile use case.
    - This includes Update Availability, allowing the franchisee to define working hours and geographic preferences.

Lead Management

- Franchisees can use View Assigned Leads, an extension of Manage Profile.
- Promising leads can be converted to clients using "Convert Lead to Client".
    - This includes updating the client's information.

Service Scheduling and Invoicing

- Once a client is created, the Franchisee uses Schedule Service to define service type, date, and time.
- Scheduling leads to Create Service, forming the basis for invoicing.
- An invoice is then generated through the Generate Invoice use case.
    - This includes the Create Service step to enforce the rule that services must exist before being invoiced.

Enhanced Patient Care

- The system supports Scan Medication via AR, allowing users to scan prescriptions.
    - This triggers AR overlays with pill identification, dosage guidance, and safety warnings.
- The View AR Exercise Guide use case projects 3D exercise models into the user's environment.
    - Assists clients with correct posture and repetition counting during rehabilitation:

The system supports enhanced patient care through *Scan Medication via AR* and ***View AR Exercise Guide*** use cases. Scan Medication via AR allows clients to scan their prescriptions using the mobile app camera, triggering AR overlays that visually identify the pill, display correct dosage instructions, and highlight warnings significantly improving medication adherence and reducing the risk of misuse (Nam & Park, 2023). View AR Exercise Guide supports clients undergoing rehabilitation by projecting

3D models of exercise routines into their real-world environment, helping them maintain correct form and count repetitions (Joshi et al., 2022).

AI Analytics Engine

- Engages with the system via Analyse Franchisee Performance.
    - Collects data on lead conversions, punctuality, and client feedback.
    - Generates performance metrics to help Horizon Health monitor franchisee effectiveness and support decision making.

Franchisee Financial Management

- The Manage Accounting use case extends Generate Invoice to support internal financial reconciliation.
    - Franchisees can access training content via Access Training (includes Update Training Material).
    - View their dues through View Monthly Invoices, which includes Issue Monthly Invoice (by the Accounts Department).

System

- Supports secure login and identity verification processes for user authentication.
- Displays assigned leads, manages data updates, and processes lead to client conversions.
- Links service scheduling with Create Service and Generate Invoice functionalities to ensure business rule compliance.
- Sends digital invoices to clients and synchronizes with external accounting systems for financial reconciliation.
- Generates reports and dashboards for stakeholders, including Marketing and AI Analytics Engine.
- Powers AR features such as Scan Medication and View Exercise Guide to support client care and engagement.

Supporting Actors

- Call Centre Administrator: Manages Transfer/Cancel Leads.
- Operations Manager: Can Restrict Leads based on qualifications.
- Marketing Manager: Accesses system insights via Generate Reports.
- Training Department: Maintains updated learning content.
- Accounts Department: Responsible for generating and issuing monthly invoices.
- External Accounting System: Engages via Integrate Accounting to handle financial compliance and audit operations.
- Client: Receives invoices through Receive Invoice and interacts passively with the system.

Assumptions
• The franchisee, as the primary user, has reliable internet access and a compatible device to perform all mobile app functions. • System-to-system communication (e.g., AI Analytics Engine, External Accounting System) is conducted securely via authenticated APIs. • All actors operate under defined roles with enforced role-based access to maintain data security and functional boundaries. • Notifications are triggered automatically through an integrated messaging service, requiring no manual input. • AI-driven performance analysis depends on the availability of accurate and consistent operational data, including service history and client feedback. • The AR functionality is internal to the app and transparent to the user, powered by an integrated AR framework (e.g., ARKit/ARCore) within the mobile app.

Table 3: Identifying Super Actors and Sub Actors

Use Case	Actors Involved
View Assigned Leads	Franchisee, Call Centre Administrator
Generate Reports	Marketing Manager, Operations Manager, AI Analytics Engine
Receive Invoice / Issue Invoices	Client, Accounts Department, Franchisee
Integrate Accounting	Franchisee, External Accounting System
Access Training	Franchisee, Training Department
Restrict Leads	Operations Manager, System
Create Service / Schedule Service	Franchisee, System
Update Profile / Availability	Franchisee, System
Convert Lead to Client	Franchisee, System

While the use case diagram presents individual actors distinctly to maintain clarity, this summary identifies shared responsibilities such as lead management, service scheduling, invoice generation, reporting, and training-related activities. For instance, both the Franchisee and Call Centre Administrator engage in lead handling, while reporting functions are performed by the Marketing Manager, Operations Manager, and AI Analytics Engine. Similarly, financial interactions are shared between the Franchisee, Accounts Department, and External Accounting System. These commonalities suggest opportunities for grouping actors under super-actor roles conceptually, though this abstraction was intentionally not shown in the visual diagram to preserve actor-specific traceability.

2. Section 2

2.1 Sequence Diagram – Lead to Invoice Process

Figure 2: Sequence diagram showing the franchisee workflow from lead to invoice

Franchisee Login and Profile Validation

The process starts when the franchisee logs into the mobile app. Upon submitting login credentials, the system calls the *Validation* Service to verify the accuracy and completeness of the franchisee's profile. This ensures that only verified users can proceed with client and invoicing operations.

Viewing Assigned Leads

Once logged in, the franchisee requests to view their *assigned leads*. The system retrieves this data from the backend and displays a list of leads, enabling the franchisee to identify and follow up with potential clients based on system assignments.

Validating and Converting the Lead

The franchisee decided to *convert lead into a client*. Before proceeding, the system validates the lead via the *Validation Service*, checking that contact details and status are accurate. Upon successful validation, the system generates a new client record linked to the franchisee.

Sending Welcome Notification

Once the client record is created, the *NotificationService* is triggered to send a welcome message. This communication confirms the client's registration and introduces the assigned franchisee.

Scheduling a Service

The franchisee then *schedules a service* by selecting the date, time, and service type. These details are saved under the client's profile, preparing the system for downstream billing operations.

Sending Service Confirmation

After scheduling, the *NotificationService* sends a confirmation message to the client. This ensures the client is informed about the upcoming appointment and its details.

Generating the Client Invoice

Following service scheduling, the system creates a *digital Invoice* for the client. This includes service type, appointment date, and payment amount, ready for external accounting processing.

Creating a Franchisee Fee Invoice

Simultaneously, a *FranchiseeInvoice* is generated for internal billing. It includes franchisee related fees such as commissions or platform charges, supporting internal financial reporting.

Pushing Invoices to External Accounting

Both the client and franchisee invoices are pushed to the *ExternalAccountingSystem*. This integration allows Horizon Health to centralize billing data for better financial tracking and compliance.

Updating Service and Client Status

Finally, the system updates the service status to “Invoiced” and the client status to “Active,” indicating that billing is complete and the client is fully onboarded. This sequential automation ensures accuracy, improves client engagement, and streamlines end to end billing through system-to-system integration (Cresswell & Sheikh , 2013)

Assumptions
• One system handles everything: We assume that the same digital platform (System/Mobile App) is used by the franchisee to manage leads, services, and invoicing. • Leads are assigned by the central system: Leads are auto-allocated to the logged-in franchisee, who can view and act on them. • Notifications are automated: Notifications like welcome messages and service confirmations are triggered by the system, not manually sent. • All invoicing is system-generated: Both client and franchisee fee invoices are created and pushed to the external accounting system through automation. • Validation is mandatory: Every lead and franchisee profile must pass backend validation before further processing.

2.2 Class Diagram – Domain Model

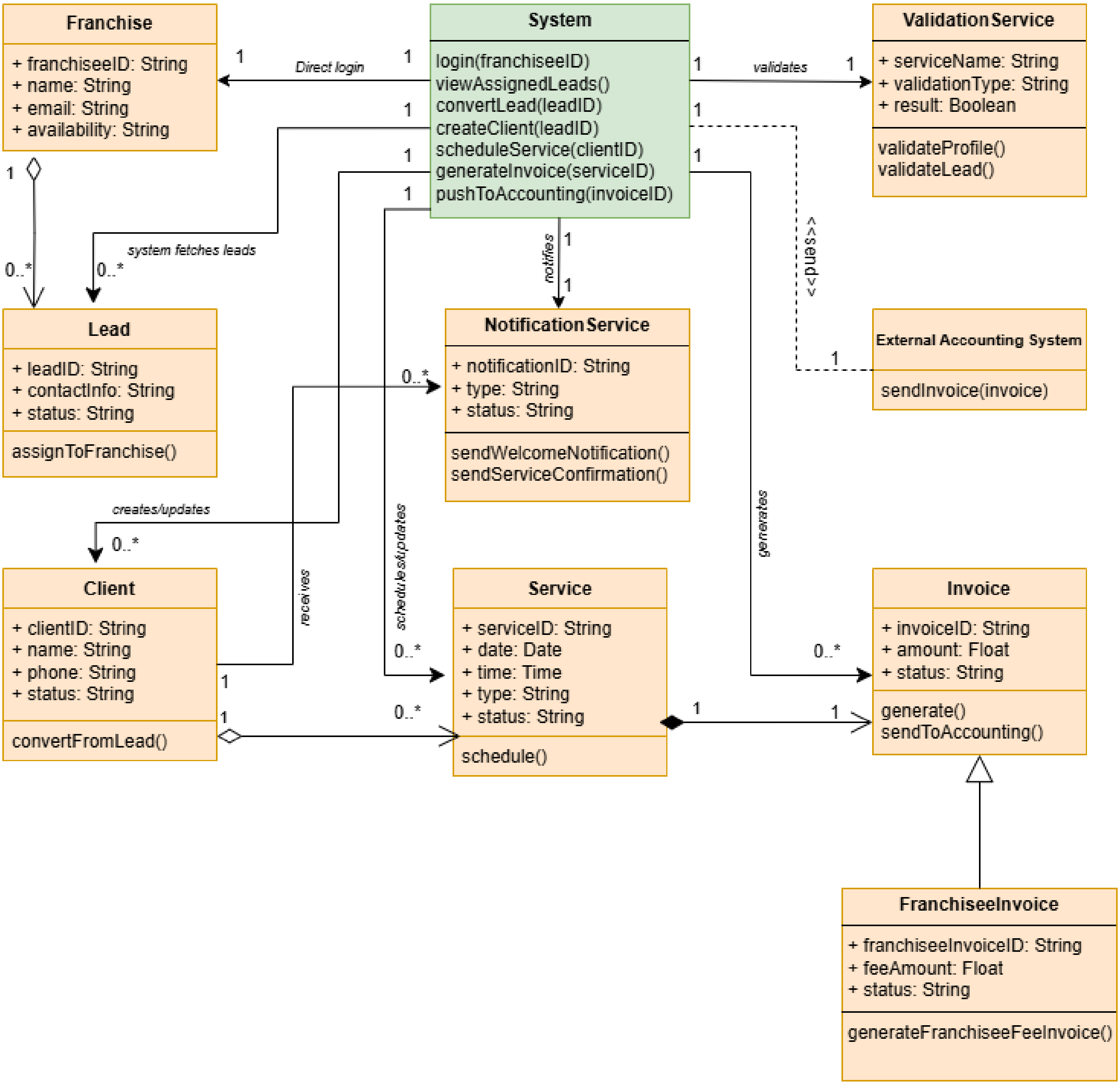


Figure 3: Class diagram representing key entities in the lead to invoice process

At the core of the model is the System (Mobile App) class, which acts as the orchestrator for franchisee operations. It handles actions such as logging in, converting leads, scheduling services, and generating invoices. Serving as a central controller, the System interacts with other classes without storing data itself. This promotes modularity and a clear separation of concerns.

The ***Franchisee*** class represents the user of the system. Each franchisee can manage multiple leads, forming a one-to-many aggregation relationship ***(1 to 0..*)***. Aggregation is used because leads exist independently and can be reassigned. Each lead, however, is only associated with one franchisee at a time.

When a ***lead*** is qualified, it is converted into a client, forming a ***1 to 1*** association. This accurately reflects the business logic where a single lead becomes exactly one client, and each client originates from one lead. The Client class includes fields such as name, contact information, and status indicators like "New" or "Active" to track progress.

A ***Client*** may be associated with multiple Service entries over time, such as consultations or follow ups. This is a ***1 to 0..**** (one to many) aggregation, as each service is linked to one client, but services can exist and be managed independently. The Service class includes a status field (e.g., "Scheduled", "Invoiced") to reflect workflow progress.

Each ***Service*** is tied to an Invoice through a ***1 to 1*** composition relationship, meaning every service results in one invoice. Composition is used because the invoice's lifecycle is dependent on the service; if the service is cancelled, the invoice is deleted.

To support internal accounting, the system includes a ***Franchisee Invoice*** class, which generalizes the ***Invoice*** class. This inheritance allows shared invoice attributes (like amount and date) while supporting franchisee specific billing logic such as service fees or reimbursements.

For validation processes, the System has a ***1 to 1*** association with a **Validation Service,** which ensures completeness and accuracy, such as checking franchisee details or lead data before conversion.

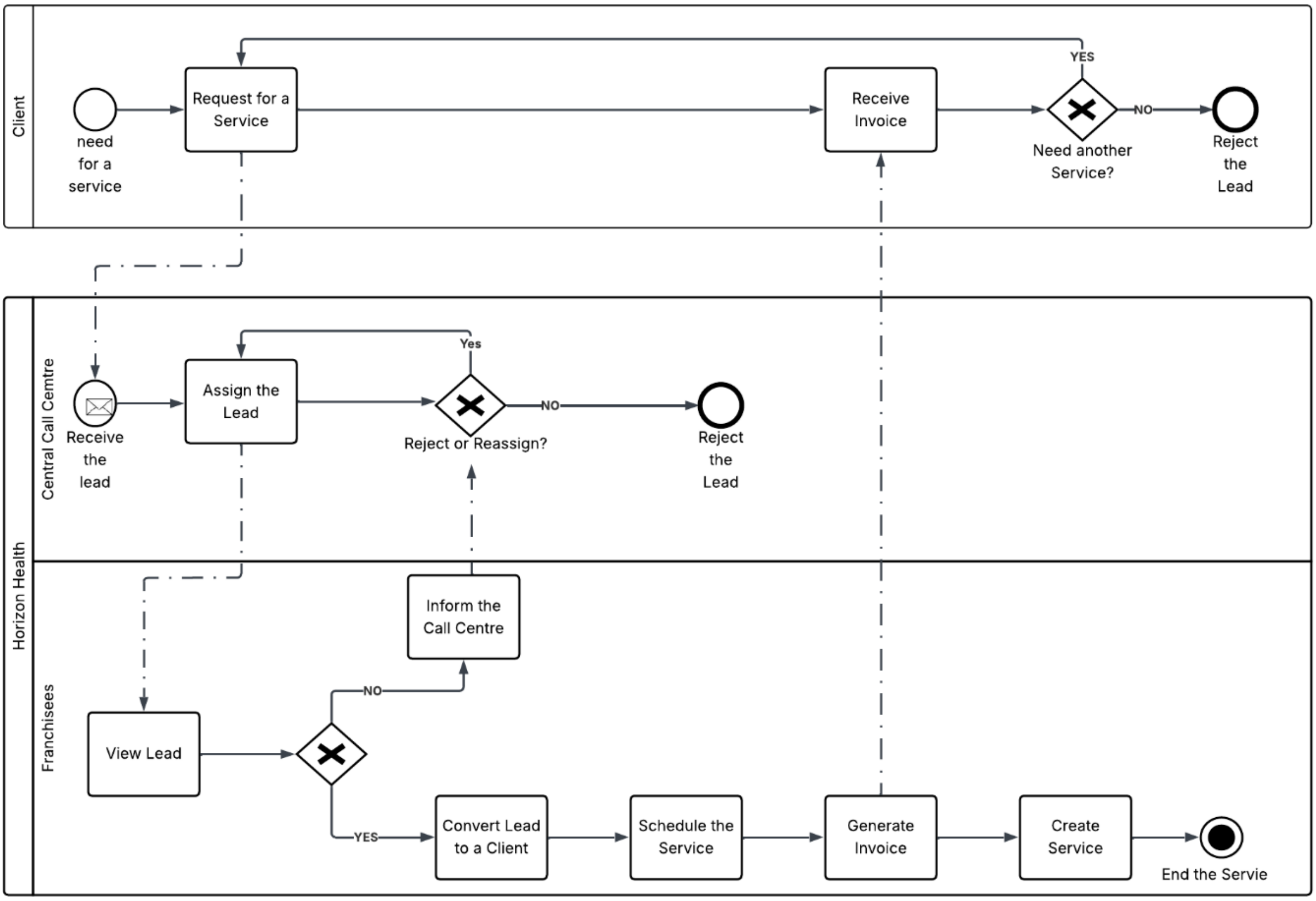
Similarly, a ***1 to 1*** link exists between the System and ***Notification Service***, which manages all outgoing communications like welcome emails or service confirmations, triggered by system events.

Lastly, the Invoice class depends on an ***External Accounting System*** via a dependency relationship. This allows the system to send finalized invoice data without creating tight coupling, supporting external integration for compliance and accounting. This object-oriented structure supports maintainability, scalability, and safe integration with external systems principles vital in modern health IT solutions (David et al., 2015)

3. Section 3

3.1 Business Process Model

The provided BPMN 2.0 diagram represents a comprehensive and well-structured model of Horizon Health's workflow for lead generation, lead cancellation or transfer, client servicing, and invoicing. The model adopts a top-down approach, where the high-level activities (such as assigning leads or scheduling services) are broken down into progressively detailed sub processes like selecting service type or generating invoices. It integrates two primary pools, Client and Horizon Health, with the latter containing two distinct lanes for Franchisees and the Central Call Centre.



Client Pool

The Client pool models the initial demand and final outcomes in the workflow. It begins with the start event "Need for a Service", followed by the task "Request for a Service". This reflects a client initiating a service request through the Horizon Health mobile platform. Once the service is rendered, the client proceeds to the "Receive Invoice" task. A key decision point follows through the exclusive gateway "Need another Service?" If the client needs further service, the process loops back to the beginning; if not, it terminates at the "Reject the Lead" end event, indicating closure of engagement.

Horizon Health Pool – Central Call Centre Lane

The Central Call Centre Lane is responsible for lead management and reassignment operations. The lane begins with the intermediate message event "Receive the Lead", triggered by the client's service request. This leads to the task "Assign the Lead", in which the call centre allocates the lead to an appropriate franchisee. If the franchisee accepts the lead, the process continues. But if Franchisees inform the call centre about their unavailability to accept the client or inform if they have reached their capacity and forgot to change their availability, call centre will come to an exclusive gateway "Reject or Reassign?" It will loop back to reassignment if there are any other franchisees available. If reassignment is not feasible, the process ends at "Reject the Lead", an end event. This pathway reflects operational handling of franchisee capacity issues or unavailability, showing that the call centre has control over lead distribution and fallback measures.

Horizon Health Pool – Franchisee Lane

This lane outlines the franchisee's end to end workflow in the Horizon Health business process. It begins with "View Lead", where the franchisee assesses an assigned lead. An exclusive gateway follows, where the franchisee decides whether to accept the lead. If the lead is deemed unsuitable, the franchisee "Informs the Call Centre", triggering a reassignment or cancellation process.

If the lead is accepted, the franchisee proceeds to "Convert Lead to a Client", a use case that includes updating the client's information such as name, contact details, and other relevant attributes.

Once the lead is successfully converted, the service delivery lifecycle begins with the franchisee initiating "Schedule Service". This step is composed of two included sub tasks, "Select Service Type" and "Set Date and Time", which together define the nature and timing of the scheduled care. These inclusions ensure that services are booked appropriately according to both client needs and franchisee availability.

After scheduling, the franchisee proceeds to "Generate Invoice", which captures billing information aligned to the scheduled service. The final step is to "Create Service", where the operational details are logged and the service becomes an actionable item in the system. The process concludes with "End the Service", symbolising the completion of the service cycle, successful delivery to the client, and backend reconciliation including updates to records and accounts. This BPMN structure ensures a seamless, role driven coordination between clients, franchisees, and administrative functions, enabling optimized lead handling, service delivery, and financial reconciliation (Islind et al., 2019)

Assumptions
• Clients initiate service via a mobile app. • Leads are initially assigned by the Call Centre, not self-selected. • Invoice can be generated before or after service depending on client needs.

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